CMSE 801 - Day 4

If statements and functions

Sep 14, 2022

General Course Announcements

- Pre-class assignments for Monday are now due Saturday at 5:30 p.m. instead of Friday
- Pre-class assignments for Wednesday are now due Tuesday at 5:30 p.m.
- Homework 1 has been released and is due Wednesday Sep 28 at 11:59 p.m.
- My slides are available on D2L
- **Late assignments will not be accepted anymore unless there are extenuating circumstances**
- Time to shuffle the groups!

Questions/comments from Day 4 PCA

- I understood this lesson and will try to help others. You may suggest to students to copy/paste error messages into google, since most of the answers can be found on stack overflow.
- Can I put a function in the list? Can a function also be the input or output of another function?
- Any recommendations for good resources of practicing coding? Because I think I need more practice.
- More function examples in class
- What if I define 2 functions which contain each other? Will it go to dead cycle?

- In general, I think I need to review my math concepts. It's been a very long time. / Trying to keep up with the mathematical aspects in the assignment / Do you have any resource recommendations for general math review?
- why would we use elif after if and not if after if?
- Is there a way to keep the variables declared inside of a function, but not necessarily returned, working through the remaining of the code after the function is called?
- please go over creation of functions again.
- nested for loops are still tricky
- Thank you for the classes!

In [1]:

```python
employees = ['Luke', 'Ningyu', 'Nate', 'Justin', 'Katrina', 'Jacob']
for e in employees:
    print(e)
    if e == "Luke" or e == "Jacob":
        print("You have to work from 9am-12pm")
```

```python
    elif (e == "Ningyu" or e == "Justin") or e == "Katrina":
        print("You have to work from 6-9pm")
    else:
        print("You have to work from 12-6pm")
```

```
Luke
You have to work from 9am-12pm
Ningyu
You have to work from 6-9pm
Nate
You have to work from 12-6pm
Justin
You have to work from 6-9pm
Katrina
You have to work from 6-9pm
Jacob
You have to work from 9am-12pm
```

In [2]:
```python
name = ['Luke', 'Ningyu', 'Nate', 'Justin', 'Katrina']
grade = [9, 10, 7, 8, -1]
info = list()
for i in range(len(name)):
    student_info = list()
    student_info.append(name[i])
    student_info.append(grade[i])
    info.append(student_info)
print(info)
```

```
[['Luke', 9], ['Ningyu', 10], ['Nate', 7], ['Justin', 8], ['Katrina', -1]]
```

In [3]:
```python
def add_info_student(exam,name,grade):
    exam.append([name,grade])
```

In [4]:
```python
name = ['Luke', 'Ningyu', 'Nate', 'Justin', 'Katrina']
grade = [9, 10, 7, 8, -1]
info = list()
for i in range(len(name)):
    add_info_student(info,name[i],grade[i])
print(info)
```

```
[['Luke', 9], ['Ningyu', 10], ['Nate', 7], ['Justin', 8], ['Katrina', -1]]
```

In [5]:
```python
def info_student(name,grade):
    exam = list()
    for i in range(len(name)):
        add_info_student(exam,name[i],grade[i])
    return exam
```

In [6]:
```python
info = info_student(name,grade)
print(info)
```

```
[['Luke', 9], ['Ningyu', 10], ['Nate', 7], ['Justin', 8], ['Katrina', -1]]
```

In [7]:
```python
def get_stat_exam(exam_info):
    nb_absence = 0
    absence_list = list()
    nb_success = 0
    nb_attend = 0
    mean = 0
    for i in exam_info:
        if i[1] > 6:
            nb_success += 1
```

```
                nb_attend += 1
                mean += i[1]
            elif i[1] == -1:
                nb_absence += 1
                absence_list.append(i[0])
            else:
                nb_attend +=1
                mean += i[1]
    mean /= nb_attend
    return mean, nb_success, nb_absence, absence_list
```

In [8]:
```
mean, nb_success, nb_absence, absence_list = get_stat_exam(info)
print(mean, ' ', nb_success, ' ', nb_absence, ' ', absence_list)
```

```
8.5   4   1   ['Katrina']
```

In [9]:
```
def get_stat_exam(exam_info):
    nb_absence = 0
    absence_list = list()
    nb_success = 0
    nb_attend = 0
    mean = 0
    for i in exam_info:
        if i[1] > 6:
            nb_success += 1
            nb_attend += 1
            mean += i[1]
        elif i[1] == -1:
            nb_absence += 1
            absence_list.append(i[0])
        else:
            nb_attend +=1
            mean += i[1]
    mean /= nb_attend
    print(mean, ' ', nb_success, ' ', nb_absence, ' ', absence_list)
```

In [10]:
```
get_stat_exam(info)
```

```
8.5   4   1   ['Katrina']
```

In [11]:
```
def add_info_student(exam,name='unknown',grade='-1'):
    exam.append([name,grade])
```

In [12]:
```
exam = list()
add_info_student(exam)
print(exam)
```

```
[['unknown', '-1']]
```

In [13]:
```
function_lists = list()
function_lists.append(add_info_student)
function_lists.append(get_stat_exam)
print(function_lists)
print(function_lists[0])
function_lists[0](exam)
print(exam)
function_lists[1](info)
```

```
[<function add_info_student at 0x7fcbc34771f0>, <function get_stat_exam at 0x7fc
bc3477ee0>]
```

```
<function add_info_student at 0x7fcbc34771f0>
[['unknown', '-1'], ['unknown', '-1']]
8.5    4    1    ['Katrina']
```

In [14]:

```
def add(x,y):
    return x+y
def multiplication(x,y):
    return x*y

def operation(function,a,b):
    return function(a,b)

print(operation(add,3.0,2.0))
print(operation(multiplication,3.0,2.0))
```

```
5.0
6.0
```